

ASSIGNMENT 6

Title: Embedded C program for timer programming ISR based buzzer on/off.

Objective : To learn interfacing of buzzer with PIC 18FXXX microcontroller

Problem statement: Write an Embedded C program for timer programming

Input:

```
#include<pic18f4550.h>
#define Buzzer LATAbits.LATA5
unsigned int count = 0;

void interrupt Timer1_ISR()
{
    if (TMR1IF==1)
    {
        //TMR1=0xCF2C;
        TMR1L = 0x20;
        TMR1H = 0xD1;
        count ++;

        if (count >= 1000)
        {
            Buzzer =~ Buzzer;
            count = 0;
        }
        TMR1IF = 0 ;
    }
}
```

```
}
```

```
void main()
```

```
{
```

```
    TRISB=0;
```

```
    TRISAbits.TRISA5 = 0;    //set buzzer pin RA5 as output
```

```
    GIE=1;
```

```
    PEIE=1;
```

```
    TMR1IE=1;
```

```
    TMR1IF=0;
```

```
/* enable 16 bit TMR1 register,no pre scale,internal clock,timer off */
```

```
    T1CON=0x80;
```

```
    TMR1L=0x20;
```

```
    TMR1H=0xD1;
```

```
    TMR1ON=1;
```

```
    while(1);
```

```
}
```

